GATE ME 90 Days Challenge

Day 60/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Weekly Revision	Week 10 Revision	42 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on a mixed revision of this week: casting (moulding processes, Chvorinov's rule, centrifugal casting), forming (work of deformation, ironing, spring-back), spatial aptitude (clock images, painted cubes, dice), joining, powder metallurgy and polymers, and machining mechanics and tool life. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

In shell moulding, the mould is made by

(A) ramming green sand around a wooden pattern

(B) dumping a mixture of fine sand and thermosetting resin onto a heated metal pattern, so that a thin cured shell forms

(C) pouring a ceramic slurry around a wax pattern

(D) blowing sodium-silicate-bonded sand into a core box and passing CO_2

Q2 NAT

A 50 mm cube casting solidifies in 90 s in a sand mould. Using Chvorinov's rule with exponent 2, the solidification time (in s, round off to one decimal place) of a cylindrical casting of 50 mm diameter and 100 mm height, poured in the same metal and the same sand, is _____

Q3 MCQ

In centrifuging (centrifuge casting), as distinct from true centrifugal and semi-centrifugal casting,

(A) the casting is always a hollow pipe with no core

(B) the mould rotates about the axis of a single large symmetric casting

(C) several small mould cavities are placed around a central sprue and the whole assembly is rotated, so the metal is forced into the cavities

(D) the mould is held stationary and only the ladle rotates

Q4 NAT

A cylindrical billet 60 mm high is upset at room temperature to a height of 40 mm. The flow curve of the material is $\sigma = 600\varepsilon^{0.25}$ MPa. The ideal work of deformation per unit volume (in MJ/m³, round off to one decimal place) is _____

Q5 MCQ

Ironing in deep drawing is used to

(A) reduce the wall thickness of a drawn cup and make it uniform, by pushing the cup through a die with a clearance smaller than the wall thickness

(B) increase the cup diameter

(C) remove the flange wrinkles from the blank before drawing

(D) form a flat blank into a shallow dish

Q6 NAT

A steel strip 2 mm thick is bent to an inside radius of 50 mm. Yield strength = 300 MPa and E = 200 GPa. Using $R_i/R_f = 4x^3 - 3x + 1$ with $x = R_iY/(Et)$, the inside radius after spring-back (in mm, round off to two decimal places) is _____

Q7 MCQ

The actual time on a wall clock is 3:40. Its image in a plane mirror held vertically beside the clock shows

(A) 9:20

(B) 8:40

(C) 8:20

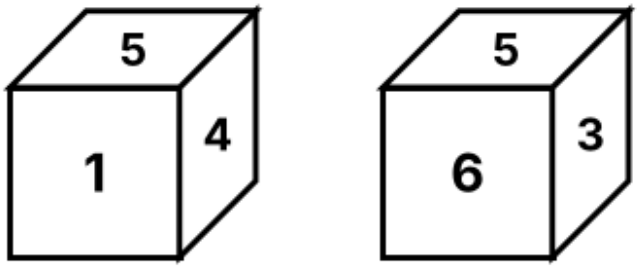
(D) 3:20

Q8 NAT

A wooden cuboid 6 cm × 5 cm × 4 cm is painted on all its faces and then cut into cubes of side 1 cm. The number of cubes with no face painted is _____

Q9 MCQ

Two views of the same die, whose faces are numbered 1 to 6, are shown. Which number is on the face opposite to 1?



(i) (ii)

(A) 3

(B) 4

(C) 2

(D) 6

Q10 MCQ

Projection welding is a variant of resistance welding in which

(A) the heat comes from a laser focused on the joint

(B) a rotating wheel electrode makes a continuous seam

(C) the current and pressure are localised at small embossed projections on one of the parts, so several welds can be made at once

(D) the two parts are butted and flashed before upsetting

Q11 NAT

Iron powder with an apparent (loose) density of 3.0 g/cm³ is compacted in a die to a green density of 6.3 g/cm³. To obtain a compact 12 mm high, the depth to which the die cavity must be filled with loose powder (in mm, round off to one decimal place) is _____

Q12 MCQ

Which one of the following statements about thermosetting polymers is correct?

(A) They can be re-melted and moulded again many times

(B) They are processed mainly by blow moulding

(C) They have linear molecular chains with no cross-links

(D) Once cured, they form a three-dimensional cross-linked network and cannot be softened by reheating

Q13 NAT

In orthogonal cutting with a tool of rake angle −6°, the cutting force is 1400 N and the thrust force is 700 N. The coefficient of friction at the tool–chip interface (round off to three decimal places) is _____

Q14 MCQ

As the cutting speed is increased from a low value when machining a ductile steel with an HSS tool, the built-up edge

(A) first grows and then disappears, because the higher temperature softens the welded layer so that the chip carries it away

(B) grows steadily larger

(C) is not affected by the cutting speed

(D) appears only above a certain high speed

Q15 NAT

For a tool with Taylor exponent n = 0.25, the percentage increase in tool life when the cutting speed is reduced by 20 % (round off to one decimal place) is _____

Answer Key

Q	Type	Answer
Q1	MCQ	(B) dumping a mixture of fine sand and thermosetting resin onto a heated metal pattern, so that a thin cured shell forms
Q2	NAT	129.6 (accept 128.9 to 130.3)
Q3	MCQ	(C) several small mould cavities are placed around a central sprue and the whole assembly is rotated, so the metal is forced into the cavities
Q4	NAT	155.3 (accept 153.8 to 156.9)
Q5	MCQ	(A) reduce the wall thickness of a drawn cup and make it uniform, by pushing the cup through a die with a clearance smaller than the wall thickness
Q6	NAT	56.32 (accept 56.04 to 56.60)
Q7	MCQ	(C) 8:20
Q8	NAT	24
Q9	MCQ	(D) 6
Q10	MCQ	(C) the current and pressure are localised at small embossed projections on one of the parts, so several welds can be made at once
Q11	NAT	25.2 (accept 25.0 to 25.4)
Q12	MCQ	(D) Once cured, they form a three-dimensional cross-linked network and cannot be softened by reheating
Q13	NAT	0.375 (accept 0.372 to 0.378)
Q14	MCQ	(A) first grows and then disappears, because the higher temperature softens the welded layer so that the chip carries it away
Q15	NAT	144.1 (accept 143.4 to 144.9)

Detailed Solutions

Q1 **Answer: (B) dumping a mixture of fine sand and thermosetting resin onto a heated metal pattern, so that a thin cured shell forms**

Concept: Shell moulding (Croning process) uses resin-bonded sand that cures on contact with a hot pattern.

The metal pattern is heated to about 200–250 °C, the sand–phenolic resin mix is dumped on it, a shell 5–10 mm thick cures, the excess is tipped off and the shell is baked: **(B)**.

Two shell halves are clamped and backed up for pouring; the result is good accuracy and finish, with a metal pattern that suits mass production.

Exam tip: (C) is investment casting and (D) is the CO_2 process.

Q2 **Answer: 129.6 (accept 128.9 to 130.3)**

Concept: $t = B(V/A)^2$; the mould constant B is found from the cube.

Cube: $V/A = a/6 = 8.333$ mm.

Cylinder: $V = \pi \times 25^2 \times 100 = 196\,350$ mm³; $A = 2\pi \times 25^2 + \pi \times 50 \times 100 = 19\,635$ mm²; $V/A = 10.0$ mm.

$t = 90 \times (10.0/8.333)^2 = 90 \times 1.44 =$ **129.6 s**

Exam tip: For a cylinder losing heat on all faces, $V/A = DH/(2D + 4H)$.

Q3 **Answer: (C) several small mould cavities are placed around a central sprue and the whole assembly is rotated, so the metal is forced into the cavities**

Concept: All three processes use rotation to force metal into the mould, but differ in the geometry.

True centrifugal: hollow parts (pipes) without cores, axis of rotation = axis of the part. Semi-centrifugal: solid symmetric parts (wheels, pulleys) with a central sprue.

Centrifuging: many small, not necessarily symmetric cavities are arranged radially around a central sprue and rotated: **(C)**.

Exam tip: In all three, the denser, cleaner metal is thrown outward, so inclusions collect near the axis.

Q4 **Answer: 155.3 (accept 153.8 to 156.9)**

Concept: Work per unit volume $= \int \sigma \, d\varepsilon = K\varepsilon^{n+1}/(n + 1)$ (the area under the flow curve).

$\varepsilon = \ln(60/40) = 0.4055$

$w = 600 \times 0.4055^{1.25}/1.25 = 600 \times 0.3235/1.25 =$ **155.3 MJ/m³** (= 155.3 N·mm/mm³)

Exam tip: 1 MPa = 1 MJ/m³, so the units come out directly.

Q5 **Answer: (A) reduce the wall thickness of a drawn cup and make it uniform, by pushing the cup through a die with a clearance smaller than the wall thickness**

Concept: In a drawn cup the wall thickens towards the rim (the flange was compressed circumferentially).

In ironing the punch pushes the cup through a die ring whose clearance is less than the wall thickness, so the wall is thinned, made uniform and lengthened: **(A)**.

Exam tip: Beverage cans are made by drawing followed by two or three ironing stages.

Q6 **Answer: 56.32 (accept 56.04 to 56.60)**

Concept: On unloading, the elastic recovery opens the bend, so the final radius R_f is larger than the die radius R_i.

$x = 50 \times 300/(200\,000 \times 2) = 0.0375$

$R_i/R_f = 4(0.0375)^3 - 3(0.0375) + 1 = 0.000211 - 0.1125 + 1 = 0.8877$

$R_f = 50/0.8877 =$ **56.32 mm**

Exam tip: Spring-back grows with Y/E and with R/t; it is compensated by over-bending, bottoming or stretch bending.

Q7 **Answer: (C) 8:20**

Concept: For a vertical mirror, image time = 11:60 − actual time (for times between 1:00 and 11:00).

11:60 − 3:40 = **8:20**: **(C)**.

Check: the minute hand at 8 (40 min) appears at 4 (20 min); the hour hand just past the 3 appears just before the 9 (between 8 and 9).

Exam tip: For actual times between 11:00 and 1:00 use 23:60 − time instead (e.g. 12:20 → 11:40).

Q8 **Answer: 24**

Concept: Unpainted cubes form the inner block, one layer in from every face: (l − 2)(b − 2)(h − 2).

(6 − 2)(5 − 2)(4 − 2) = 4 × 3 × 2 = **24**

Check: total 120; with one face painted 2[(4)(3) + (4)(2) + (3)(2)] = 52; with two faces 4(4 + 3 + 2) = 36; corners 8. 24 + 52 + 36 + 8 = 120.

Exam tip: Always check that the four groups add up to the total number of cubes.

Q9 **Answer: (D) 6**

Concept: When the same face is on top in two views, the side faces seen in the two views belong to one ring of four faces around it.

5 is on top in both views, so 1, 4, 6 and 3 are the four side faces around 5 (and 2 is opposite 5).

Looking down on 5, the side faces in clockwise order are front, left, back, right. View (i) gives right 4 → front 1, and view (ii) gives right 3 → front 6. The only cycle containing both steps is 4 → 1 → 3 → 6.

Faces two places apart in the ring are opposite: 1 is opposite **6** (and 4 is opposite 3): **(D)**.

Exam tip: A face can never be opposite a face that appears next to it in any view; here only 6 and 3 were possible, and the order of the faces decides.

Q10 **Answer: (C) the current and pressure are localised at small embossed projections on one of the parts, so several welds can be made at once**

Concept: In resistance welding, heat H = I^2Rt is concentrated where the contact area (and so the current density) is small.

In projection welding, dimples or projections pressed into one sheet localise the current; large flat electrodes then make several welds in one stroke: **(C)**.

(B) is seam welding and (D) is flash butt welding.

Exam tip: Weld nuts and studs are attached to car body panels by projection welding.

Q11 **Answer: 25.2 (accept 25.0 to 25.4)**

Concept: The mass is the same before and after pressing, and the cross-section is fixed by the die, so the height ∝ 1/density.

H_{fill} × 3.0 = 12 × 6.3

H_{fill} = 12 × 6.3/3.0 = **25.2 mm**

Exam tip: The fill-height ratio (compression ratio) 2.1 here sets the die depth and the punch stroke.

Q12 **Answer: (D) Once cured, they form a three-dimensional cross-linked network and cannot be softened by reheating**

Concept: Curing (by heat, catalyst or radiation) forms covalent cross-links between the chains.

The network does not melt on reheating; it chars or decomposes instead: **(D)**.

Thermoplastics have linear or branched chains held by weak secondary bonds, so they soften on heating and can be recycled.

Exam tip: Thermosets are shaped by compression or transfer moulding and reaction injection moulding; thermoplastics by injection, extrusion and blow moulding.

Q13 **Answer: 0.375 (accept 0.372 to 0.378)**

Concept: $F = F_c \sin\alpha + F_t \cos\alpha$ and $N = F_c \cos\alpha - F_t \sin\alpha$, with α taken with its sign.

$\sin(-6°) = -0.1045$, $\cos 6° = 0.9945$

$F = 1400(-0.1045) + 700(0.9945) = -146.3 + 696.2 = 549.8$ N

$N = 1400(0.9945) - 700(-0.1045) = 1392.3 + 73.2 = 1465.5$ N

$\mu = 549.8/1465.5 =$ **0.375**

Exam tip: With a negative rake, part of F_c acts against the friction force, so ignoring the sign of α gives a badly wrong μ (here 0.639 instead of 0.375).

Q14 **Answer: (A) first grows and then disappears, because the higher temperature softens the welded layer so that the chip carries it away**

Concept: The BUE forms when chip material work-hardens and welds to the rake face at moderate temperatures.

At very low speeds it grows and periodically breaks off; as the speed (and interface temperature) rises, the welded layer softens and is swept away, so it disappears: **(A)**.

Exam tip: This is why surface roughness often improves sharply once the cutting speed exceeds the BUE range.

Q15 **Answer: 144.1 (accept 143.4 to 144.9)**

Concept: $VT^n = C$ gives $T \propto V^{-1/n} = V^{-4}$.

$T_2/T_1 = (1/0.8)^4 = 1.25^4 = 2.4414$

Increase = **144.1 %**

Exam tip: Small n means tool life is very sensitive to speed: a 20 % speed cut more than doubles the life.

What next?

Day 60 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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